SQUINT

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Squint = misalignment of eyes. Common SAQ + clinical posting topic!

Squint | Strabismus | Ophthalmology | 3rd Year

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Classification + Types of Squint

Squint = 2-4% of kids!
Early detection saves
vision from amblyopia!

DEFINITION

Strabismus (Squint) = misalignment of the visual axes. One eye deviates while the other fixates.

Types by Direction of Deviation

Type	Direction	Key Details
Esotropia	Inward (convergent)	Most common squint in children
Exotropia	Outward (divergent)	More common in adults
Hypertropia	Upward	Usually paralytic
Hypotropia	Downward	Often floor fracture / IO palsy

Main Classification

Category	Types
Concomitant	Angle same in all gazes — infantile, accommodative, intermittent
Incomitant	Angle varies with gaze — paralytic (CN III, IV, VI palsy)
Pseudo-squint	Epicanthus, wide IPD — appears squint but not truly misaligned
Latent (Phoria)	Deviation only when fusion disrupted (cover test reveals)
Manifest (Tropia)	Constant deviation even with both eyes open

- Accommodative esotropia: Hypermetropia -> excess accommodation -> excess convergence -> esotropia
- Treatment: Full correction of hypermetropia with glasses (+lenses). Bifocals if high AC/A ratio.
- Partially accommodative: Glasses correct part; residual deviation may need surgery

This comparison table =
THE most asked question!
Learn it cold, Sammy!

Concomitant vs Paralytic (Incomitant) Squint

Feature	Concomitant	Paralytic
Angle of deviation	Same in all directions	Varies with gaze direction
Age of onset	Childhood	Any age (often adult)
Cause	Refractive, sensory, genetic	Nerve palsy (CN III, IV, VI)
Diplopia	Usually absent (suppression)	Present (adult onset)
Head posture	Usually absent	Compensatory (face turn/tilt)
Eye movements	Full in all directions	Restricted in direction of palsy
Primary deviation	= Secondary deviation	< Secondary deviation
Treatment	Glasses, patching, surgery	Treat cause, prisms, surgery

Cranial Nerve Palsies + Squint Patterns

Nerve	Muscle Affected	Clinical Features
CN III	SR, IR, MR, IO, LPS	Eye down & out, ptosis, dilated pupil
CN IV	Superior oblique	Vertical diplopia, head tilt to opposite side
CN VI	Lateral rectus	Convergent squint, face turn to affected side

★ **Primary deviation: angle when normal eye fixates. Secondary: angle when paretic eye fixates.**
In paralytic squint: secondary deviation > primary deviation (Hering's law).
In concomitant squint: primary = secondary deviation.

AC/A Ratio (Accommodative Convergence/Accommodation)

Normal AC/A ratio: 3-5 prism diopters per diopter of accommodation
High AC/A ratio: more convergence for near -> esotropia greater at near -> Rx: bifocals
Methods: Heterophoria method (calculated), Lens gradient method (clinical)

Diagnostic Tests for Squint

Cover test + Hirschberg
= clinical posting must!
Practice on friends!

Hirschberg Test (Corneal Light Reflex)

1. Shine torch at 33cm — observe corneal light reflex position
2. Normal: reflex at center of pupil in both eyes
3. 1mm displacement = ~7 degrees (~15 prism diopters) of deviation
4. Useful in infants and uncooperative patients (quick screening test)

Cover Tests — Types + Interpretation

Test	What It Detects	Interpretation
Cover-uncover	Manifest squint (tropia)	Covered eye moves to fixate on uncover
Alternate cover	Total deviation (tropia+phoria)	Breaks fusion, reveals latent deviation
Prism cover test	Measures deviation	Prism neutralizes the movement

Other Important Tests

Test	Purpose
Krimsky test	Prism + corneal reflex — measures angle in uncooperative pts
Maddox rod	Dissociates eyes — measures heterophoria
Synoptophore	Measures angle, assesses binocular vision, stereopsis
Worth 4-dot test	Tests fusion and suppression (near + distance)
Titmus / TNO test	Tests stereopsis (depth perception)
Forced duction test	Differentiates restrictive from paralytic (under anesthesia)
Parks 3-step test	Identifies which cyclovertical muscle is paralyzed

★ Parks 3-Step Test
Step 1: Which eye is hypertropic? Step 2: Worse in right or left gaze?
Step 3: Worse on right or left head tilt? -> Identifies the paretic SO/IO muscle.

Amblyopia = lazy eye.
Treatable only in kids!
Critical period < 7-8 yrs!

Amblyopia (Lazy Eye)

Feature	Detail
Definition	Reduced VA in one/both eyes without organic cause
Mechanism	Cortical suppression during visual development
Critical period	Birth to 7-8 years (sensitive period for treatment)
Types	Strabismic, anisometropic, stimulus deprivation, meridional
Most common cause	Strabismus (strabismic amblyopia)
Diagnosis	2-line difference in VA between eyes after correction

Types of Amblyopia

Type	Cause & Features
Strabismic	Constant unilateral squint -> suppression of deviating eye
Anisometropic	Large difference in refractive error between eyes (>1.5D)
Stimulus deprivation	Congenital cataract, ptosis -> worst prognosis, needs URGENT Rx
Meridional	Uncorrected astigmatism -> amblyopia in one meridian

Treatment of Amblyopia

Remove cause:	Correct refractive error, remove cataract, fix ptosis
Occlusion therapy:	Patch the GOOD eye to force amblyopic eye to work
Penalization:	Atropine drops in good eye (alternative to patching)
Duration:	Part-time patching: 2-6 hrs/day depending on severity

Surgical Management of Squint

Procedure	Detail
Recession	Weakening procedure — move muscle insertion posteriorly
Resection	Strengthening procedure — shorten muscle by removing portion
Indication	After correction of amblyopia + glasses; residual deviation
Timing	After 6 months of stable alignment / failed conservative Rx

Squint = clinical posting
exam favorite + viva Q!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

1. Differentiate concomitant and paralytic squint. Describe management
2. Describe amblyopia — types, diagnosis, and treatment
3. Describe diagnostic tests for squint (Cover test, Hirschberg, etc.)
4. Describe accommodative esotropia — pathogenesis and management

Most Repeated SAQs (5–6 marks)

1. Cover test — types and interpretation
2. Hirschberg test — method and grading
3. AC/A ratio — definition and significance
4. CN III palsy — clinical features
5. Parks 3-step test
6. Occlusion therapy for amblyopia

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common squint in children	Esotropia (convergent)
Eye down and out + ptosis + mydriasis	CN III palsy (complete)
Head tilt test (Bielschowsky)	Superior oblique palsy (CN IV)
Face turn toward paretic side	CN VI (lateral rectus) palsy
Screening test for squint	Hirschberg test (corneal reflex)
Gold standard for measuring deviation	Prism + alternate cover test
Pseudosquint cause	Epicanthus (wide nasal bridge)
Critical period for amblyopia Rx	< 7-8 years of age
Treatment of amblyopia	Patch the GOOD eye (occlusion)
Recession = weakening, Resection =	Strengthening procedure
Normal AC/A ratio	3-5 PD per diopter

♥ **Every child you screen for squint could be saved from amblyopia, Sammy!**
Early detection = lifelong good vision. You're going to be amazing!